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Characteristic analysis of Otsu threshold and its applications

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ABSTRACT

This paper proves that Otsu threshold is equal to the average of the mean levels of two classes partitioned by this threshold. Therefore, when the within-class variances of two classes are different, the threshold biases toward the class with larger variance. As a result, partial pixels belonging to this class will be misclassified into the other class with smaller variance. To address this problem and based on the analysis of Otsu threshold, this paper proposes an improved Otsu algorithm that constrains the search range of gray levels. Experimental results demonstrate the superiority of new algorithm compared with Otsu method.

1. Introduction

In image segmentation, thresholding becomes an effective tool to separate the object from the background when the gray levels are substantially different between them (Sezgin and Sankur, 2004; Sahoo et al., 1988). The classical threshold segmentation algorithms include histogram shape-based methods (Rosenfeld and Torre, 1983), clustering-based methods (Otsu, 1979; Kittler and Illingworth, 1985), mutual information methods (Kapur et al., 1985; Abutaleb, 1989), attribute similarity-based methods (Tsai, 1985), local adaptive segmentation methods (Sauvola and Pietaksinen, 2000), etc. Among them, Otsu method has received more attention. Several improved versions of Otsu method have been proposed, such as recursive Otsu method (Cheriet et al., 1998), 2-dimensional Otsu method (Gong et al., 1998), two-stage multi-threshold Otsu method (Huang and Wang, 2009) and so on.

Although Otsu method has been widely studied, the problems which kind of images Otsu method is suitable for and which class Otsu threshold biases toward are still open in theory. Up to now, most analysis for Otsu threshold mainly depends on experiments, however, the lack of theory analysis and the insufficiency of test images often arrive at inaccurate conclusions. For instance, when the sizes of object and background are unequal, Qiao et al. (2007) believed that Otsu threshold tends to split the larger class to make two classes keep similar size. On the contrary, Medina and Madrid (2008) thought that Otsu threshold further amplify the size difference between the object and background. The above two different opinions indicate that it is difficult to interpret the tendency of Otsu threshold only according to the sizes of two classes. In addition, Hou et al. (2006) pointed out that Otsu method tends to shift the threshold toward the component with larger size or larger class variance. However, this conclusion cannot explain the bias of Otsu threshold when the component with larger class variance has fewer pixels.

This paper discovers an important property of Otsu threshold, namely, Otsu threshold biases toward the component with larger within-class variance. This discovery can reasonably explain the experimental results of the references (Medina and Madrid, 2008; Hou et al., 2006). Furthermore, this paper discusses the relationship between the iterative threshold algorithm and Otsu method, which reveals that the thresholds calculated by the two methods are equivalent under certain condition. For an image with significant difference between within-class variances of the object and background, the property of Otsu threshold provides a theoretical basis to limit the search range of the optimal threshold.

The remainder of this paper is organized as follows. Section 2 gives a review of Otsu algorithm. Section 3 reveals an important property of Otsu threshold. Section 4 discusses the relationship between the iterative threshold algorithm and Otsu method. An improved Otsu algorithm, which constrains the search range of the optimal threshold, is proposed in Section 5. Finally, conclusions are given in Section 6.

2. Review of Otsu method

Suppose that the pixels in a given image be represented in L gray levels $(1,2,\ldots,L)$. Let n_i denote the number of pixels at level

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i, and N denote the total number of pixels, $N = \sum_{i=1}^{L} n_i$. The probability of occurrence of level i is given by $p_i = n_i/N$.

Let an image be divided into two classes C_0 and C_1 by threshold T. C_0 consists of pixels with levels $[1, \cdots, T]$ and C_1 consists of pixels with levels $[T+1, \cdots, L]$. Let $P_0(T)$ and $P_1(T)$ denote the cumulative probabilities, $\mu_0(T)$ and $\mu_1(T)$ denote the mean levels, and $\sigma_0^2(T)$ and $\sigma_1^2(T)$ denote the variances of the classes C_0 and C_1, respectively. These values are given by:

$$P_0(T) = \sum_{i=1}^{T} p_i \tag{1}$$

$$P_1(T) = \sum_{i=T+1}^{L} p_i = 1 - P_0(T) \tag{2}$$

$$\mu_0(T) = \sum_{i=1}^{T} i \frac{p_i}{P_0(T)} = \frac{1}{P_0(T)} \sum_{i=1}^{T} i p_i \tag{3}$$

$$\mu_1(T) = \sum_{i=T+1}^{L} i \frac{p_i}{P_1(T)} = \frac{1}{P_1(T)} \sum_{i=T+1}^{L} i p_i \tag{4}$$

$$\sigma_0^2(T) = \sum_{i=1}^{T} (i - \mu_0(T))^2 \frac{p_i}{P_0(T)} \tag{5}$$

$$\sigma_1^2(T) = \sum_{i=T+1}^{L} (i - \mu_1(T))^2 \frac{p_i}{P_1(T)} \tag{6}$$

Let μ, $\sigma_b^2(T)$, and $\sigma_w^2(T)$ represent the mean level of the image, the between-class variance, and the within-class variance, respectively:

$$\mu = \sum_{i=1}^{L} i p_i = P_0(T)\mu_0(T) + P_1(T)\mu_1(T) \tag{7}$$

$$\sigma_b^2(T) = P_0(T)(\mu_0(T) - \mu)^2 + P_1(T)(\mu_1(T) - \mu)^2 \tag{8}$$

$$\sigma_w^2(T) = P_0(T)\sigma_0^2(T) + P_1(T)\sigma_1^2(T) \tag{9}$$

The threshold decided by maximizing the between-class variance proposed in Otsu is:

$$T^* = \underset{1 \leqslant T < L}{\arg} \, max\{\sigma_b^2(T)\} \tag{10}$$

This value is equal to the threshold decided by minimizing the within-class variances criterion:

$$T^* = \underset{1 \leqslant T < L}{\arg} \, min\{\sigma_w^2(T)\} \tag{11}$$

Furthermore, the above threshold is same as the threshold calculated by maximizing the ratio of between-class variance to within-class variances (Sahoo et al., 1988).

3. Analysis of the property of Otsu threshold

In this section, we will proof an important property of Otsu threshold, i.e., Otsu threshold is equal to the average of the mean levels of two classes partitioned by this threshold.

Proposition 1. *Let $\mu_0(T_0)$ and $\mu_1(T_0)$ be the mean levels of two classes divided by a given threshold T_0, and $T_m = \lfloor(\mu_0(T_0) + \mu_1(T_0))/2\rfloor$, then $\sigma_w^2(T_0) \geqslant \sigma_w^2(T_m)$.*

Proof. Assume that the mean levels of two classes divided by threshold T_m are denoted by $\mu_0(T_m)$ and $\mu_1(T_m)$. □

Obviously, $\sigma_w^2(T_m) = \sigma_w^2(T_0)$ when $T_m = T_0$.
Let's assume $T_m > T_0$.
Then, for $i \in [T_0 + 1, ..., T_m]$,

$$0 < i - \mu_0(T_0) \leqslant \mu_1(T_0) - i$$

i.e., $(i - \mu_1(T_0))^2 \geqslant (i - \mu_0(T_0))^2$

According to the method of calculating the mean level, we have:

$$\sum_{i=1}^{T_m}(i - \mu_0(T_m))p_i = 0 \qquad \sum_{i=T_m+1}^{L}(i - \mu_1(T_m))p_i = 0$$

$$\begin{aligned}
\sigma_w^2(T_0) &= \sum_{i=1}^{T_0}(i-\mu_0(T_0))^2 p_i + \sum_{i=T_0+1}^{L}(i-\mu_1(T_0))^2 p_i \\
&= \sum_{i=1}^{T_0}(i-\mu_0(T_0))^2 p_i + \sum_{i=T_0+1}^{T_m}(i-\mu_1(T_0))^2 p_i + \sum_{i=T_m+1}^{L}(i-\mu_1(T_0))^2 p_i \\
&\geqslant \sum_{i=1}^{T_0}(i-\mu_0(T_0))^2 p_i + \sum_{i=T_0+1}^{T_m}(i-\mu_0(T_0))^2 p_i + \sum_{i=T_m+1}^{L}(i-\mu_1(T_0))^2 p_i \\
&= \sum_{i=1}^{T_m}(i-\mu_0(T_0))^2 p_i + \sum_{i=T_m+1}^{L}(i-\mu_1(T_0))^2 p_i \\
&= \sum_{i=1}^{T_m}(i-\mu_0(T_m))^2 p_i + \sum_{i=1}^{T_m}(\mu_0(T_m)-\mu_0(T_0))^2 p_i \\
&\quad +2(\mu_0(T_m)-\mu_0(T_0))\sum_{i=1}^{T_m}(i-\mu_0(T_m))p_i \\
&\quad + \sum_{i=T_m+1}^{L}(i-\mu_1(T_m))^2 p_i + \sum_{i=T_m+1}^{L}(\mu_1(T_m)-\mu_1(T_0))^2 p_i \\
&\quad +2(\mu_1(T_m)-\mu_1(T_0))\sum_{i=T_m+1}^{L}(i-\mu_1(T_m))p_i \\
&= \sum_{i=1}^{T_m}(i-\mu_0(T_m))^2 p_i + \sum_{i=T_m+1}^{L}(i-\mu_1(T_m))^2 p_i \\
&\quad +(\mu_0(T_m)-\mu_0(T_0))^2\sum_{i=1}^{T_m}p_i + (\mu_1(T_m)-\mu_1(T_0))^2\sum_{i=T_m+1}^{L}p_i \\
&= \sigma_w^2(T_m) + (\mu_0(T_m)-\mu_0(T_0))^2\sum_{i=1}^{T_m}p_i + (\mu_1(T_m)-\mu_1(T_0))^2\sum_{i=T_m+1}^{L}p_i \\
&\geqslant \sigma_w^2(T_m)
\end{aligned}$$

For the case of $T_m < T_0$, the certification process is similar to the above.

Proposition 2. *Assume that only one threshold value T_* makes the smallest within-class variance, i.e., for any $1 \leqslant T < L$, $T \neq T_*$, there is $\sigma_w^2(T_*) < \sigma_w^2(T)$. Let $\mu_0(T_*)$ and $\mu_1(T_*)$ be the mean levels of two classes divided by the threshold T_* respectively, and let $T_m = \lfloor(\mu_0(T_*) + \mu_1(T_*))/2\rfloor$. Then, there must be $T_* = T_m$.*

Proof. Obviously, T_* is a special case of any given value T_0 in the Proposition 1. So, the inequality $\sigma_w^2(T_*) \geqslant \sigma_w^2(T_m)$ can be drawn from the Proposition 1. □

If $T_m \neq T_*$, then exist T_m which is not equal to T_*, satisfied $\sigma_w^2(T_m) \leqslant \sigma_w^2(T_*)$. It contradicts the conditions of this Proposition.
This completes the proof.

Proposition 3. *Assume threshold T_* makes the minimum within-class variance, i.e., $\sigma_w^2(T_*) \leqslant \sigma_w^2(T)$ for all $1 \leqslant T < L$. Let $T_m = \lfloor(\mu_0(T_*) + \mu_1(T_*))/2\rfloor$. Then, one of the following three cases must be set up:*

(1) $T_* = T_m$.
(2) *If $T_* < T_m$, then, for any $i \in [T_* + 1, ..., T_m]$, $p_i = 0$.*
(3) *If $T_* > T_m$, then, for any $i \in [T_m + 1, ..., T_*]$, $p_i = 0$.*

Proof. We can see from the proof of Proposition 1 that:

$$\sigma_w^2(T_*) \geqslant \sigma_w^2(T_m) + (\mu_0(T_m) - \mu_0(T_*))^2 \sum_{i=1}^{T_m} p_i + (\mu_1(T_m) - \mu_1(T_*))^2 \sum_{i=T_m+1}^{L} p_i \tag{12}$$

To prove above conclusions we need to consider the following three cases:

(1) $T_* = T_m$, Obviously, the Proposition holds.

(2) $T_* < T_m$, if exist $p_i \neq 0$ for $i \in [T_* + 1, \ldots, T_m]$, then:

$$\begin{aligned}\mu_0(T_m) &= \left(\sum_{i=1}^{T_m} ip_i\right) \Big/ \left(\sum_{i=1}^{T_m} p_i\right) \\ &= \left(\mu_0(T_*)P(T_*) + \sum_{i=T_*+1}^{T_m} ip_i\right) \Big/ \left(\sum_{i=1}^{T_m} p_i\right) \\ &> \left(\mu_0(T_*)P(T_*) + \sum_{i=T_*+1}^{T_m} \mu_0(T_*)p_i\right) \Big/ \left(\sum_{i=1}^{T_m} p_i\right) = \mu_0(T_*)\end{aligned}$$

Similarly, the conclusion $\mu_1(T_m) > \mu_1(T_*)$ holds.

Then, $\sigma_w^2(T_*) > \sigma_w^2(T_m)$, since the last two components located in the right side of formula (12) are positive. It contradicts the conditions of this Proposition. So, for all $i \in [T_* + 1, \ldots, T_m]$, $p_i = 0$.

(3) $T_* > T_m$, the certification process is similar to the above.

This completes the proof. □

Obviously, when there is no pixels in the range of T_* and T_m, the mean levels $\mu_0(T_m)$ and $\mu_1(T_m)$ divided by T_m meet the following condition:

$$\mu_0(T_m) = \mu_0(T_*), \mu_1(T_m) = \mu_1(T_*)$$

So, $T_m = \lfloor(\mu_0(T_*) + \mu_1(T_*))/2\rfloor = \lfloor(\mu_0(T_m) + \mu_1(T_m))/2\rfloor$.

That is to say, the threshold T_m is equal to the average of the mean levels of two classes partitioned by itself. The segmentation results of threshold T_m and T_* are the same. So, the three cases in Proposition 3 can be summarized as that the optimal threshold calculated by minimizing the sum of within-class variances is the average of the mean levels of two classes.

In many applications of image segmentation, the variance of the object is substantially different from that of background, and the mean levels of the object and background are far apart. Thresholding then becomes an effective tool to separate object from the background. Such images are widespread in practical applications, such as cell images, gradient images, and the difference images of change monitoring (Hu et al., 2006). The intersection point of the distributions of the object and background is the ideal threshold to segment these images. However, this intersection is far away from the mean level of the class with large variance, and near the mean level of the class with small variance. Therefore, it is not equal to the average value of the centers of two classes. At the same time, it is also impossible to be the Otsu threshold. Furthermore, The equality of the distances from Otsu threshold to the centers of two classes indicates that the variances of two classes segmented by Otsu threshold are quite close. To this end, it will split the class with larger variance, i.e., a part of pixels of the class with larger variance will be divided into the class with smaller variance. Fig. 1 gives an example to exhibit intuitively the above description. It is seen that Otsu threshold remarkably deviates from the intersection point toward the center of the class with larger variance.

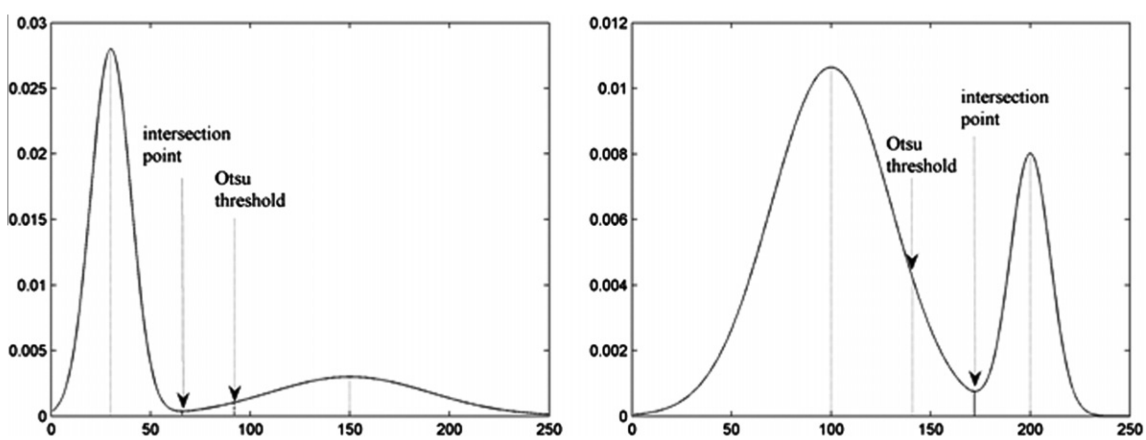

Fig. 1. Otsu threshold for segmentation of two classes with different variances.

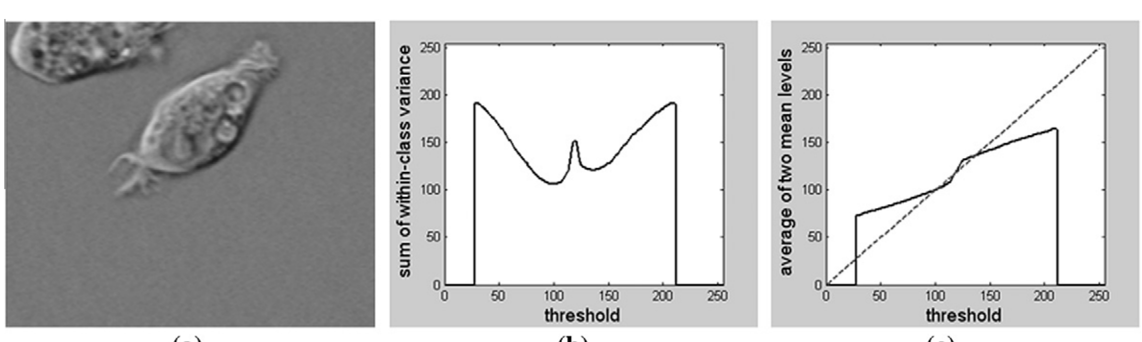

Fig. 2. Curves of the sum of within-class variance and the average of mean levels of the classes. (a) cell image, (b) curve of the sum of within-class variances, (c) curve of the average of mean levels of two classes.

Using the above property of Otsu threshold, it is easy to explain the conclusions given by Medina and Madrid (2008) and Hou et al. (2006). Medina and Madrid pointed out that Otsu threshold splits the class with more pixels to make the pixel numbers in two classes approach. However, Hou et al. (2006) gave the contrary conclusion. Observation of the test images given in the two papers finds that the variance of the class with more pixels is larger in the former and contrary in the latter. Overall, Otsu threshold is suitable for segmentation of classes with approximate or equal within-class variances.

4. Relationship between the iterative threshold method and Otsu method

For binary classification, the iterative method (Ridler and Calvard, 1978) is described as follows.

Step 1: given an initial threshold T_i, $i = 1$.
Step 2: The image is divided into two classes by threshold T_i. Calculate the mean levels of two classes: $\mu_0(T_i)$, $\mu_1(T_i)$.
Step 3: $T_{i+1} = \lfloor(\mu_0(T_i) + \mu_1(T_i))/2\rfloor$.
Step 4: if $Ti_{+1} = T_i$, terminate the algorithm. Otherwise, $i \leftarrow i + 1$, go to Step 2.

When the iterative algorithm stops, the calculated threshold is the average of the mean levels of two classes.

Dong et al. (2008) considered that the iterative method is mathematically equivalent to the Otsu method. However, this view is not correct. The threshold calculated by the iterative method may be a local extreme value of the sum of the within-class variances, and it is not necessarily equivalent to Otsu threshold.

For instance, for the image of cell.tif in MATLAB as shown in Fig. 2(a), the result of the iterative threshold method changes with different initial thresholds as shown in Table 1. However, Otsu threshold for this image is at gray level 99 certainty.

To explain above phenomenon intuitively, Fig. 2(b) gives the curve of the sum of within-class variances. The abscissa indicates the thresholds of the segmentation, whereas the ordinate indicates the sum of two within-class variances. From this figure, it is clear that there are several local minimum points in the curve. Fig. 2(c) is the curve of the average value of the mean levels of two classes. The abscissa indicates the thresholds of the segmentation, whereas the ordinate indicates the average value of mean levels of two classes. From the figure, we can see that there are several points whose thresholds are equal to corresponding average values of mean levels. These points represent the local extreme points of the sum of within-class variances. Obviously, if the initial threshold value is just one of these points, the iterative process will terminate immediately. That's to say, threshold calculated by the iterative method is one of the local extreme value of the sum of within-class variances, but not necessarily the global minimum.

From the proofs of the Propositions 1 and 3, we can see that the sum of within-class variances decreases monotonically in the iteration process. Suppose that the algorithm stops after n iterations, then, $\sigma_w^2(T_1) > \sigma_w^2(T_2) > \sigma_w^2(T_3) > \ldots > \sigma_w^2(T_{n-1}) = \sigma_w^2(T_n)$. Besides, it also can be proved that the threshold obtained in the iterative process changes monotonically. If $T_1 < T_2$, then, $T_1 < T_2 < T_3 < \ldots < T_{n-1} = T_n$, or when $T_1 > T_2$, $T_1 > T_2 > T_3 > \ldots > T_{n-1} = T_n$. When the curve of the sum of within-class variances is a unimodal function, the threshold of the iterative method is equal to Otsu threshold.

Table 1
The initial thresholds and results of iterative threshold method.

Initial thresholds	Results of the iterative method
$T \in [29, 99]$	99
$T \in [100, 118]$	100
$T = 119$	119
$T \in [120, 211]$	136

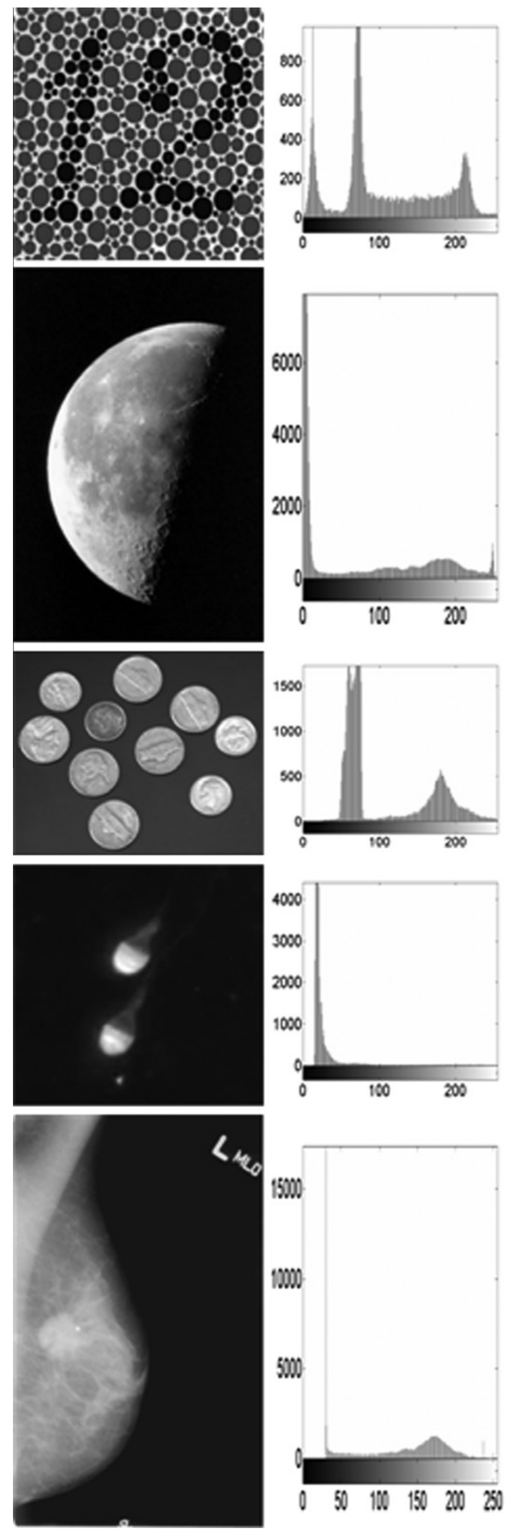

Fig. 3. Test images and their histograms. From top to bottom: color vision image, moon image, coin image, sperm image, and mammogram.

5. Improvement to Otsu method

In image segmentation, specific algorithms can be designed according to the prior knowledge of the background to enhance the quality of segmentation (Rosin and Paul, 2001). According to the property of Otsu threshold and the gray-level distributions of the object and background, an optimal threshold can be calculated. Suppose that we know that the difference of the variances of object and background is significant, and the object is brighter and its variance is larger. Then, Otsu threshold value will bias toward the object. Therefore, we can search within a limited range for the optimal threshold because it is actually smaller than Otsu threshold.

Based on the above analysis, a novel range-constrained Otsu method is proposed as follows:

Algorithm: range-constrained Otsu method

Step 1: calculate the threshold T_1 by Otsu method in the whole image.
Step 2: calculate the threshold T_2 by Otsu method in the pixels with gray levels in $[1, T_1]$.
Step 3: the pixels whose gray levels are larger than T_2 are classified as object and the others as background.

To demonstrate the efficiency of the proposed method, we compare our method with other four methods, entropic thresholding (Kapur, 1985), histogram shape-based thresholding (Rosin, 2001), non-parametric method (Medina-Carnicer, 2008), and Otsu (Otsu, 1979). The test images are the color vision image (Kwon, 2004), moon image, coin image, sperm image, and mammogram. These test images and their histograms are shown in Fig. 3. Fig. 4 shows the results of our method and the comparative methods. The first column to the last column is the results by the methods of Kapur, Rosin, Medina-Carnicer, Otsu and ours, respectively. Table 2 lists the corresponding thresholds of these methods. From the above results, it can be seen that our method succeeded as Rosin (2001) for images with unimodal histograms, which is a special case of the variance of the object substantially different from that of background. However, our method can also obtain good results for the images with non-unimodal histograms, such as color vision image.

We also use the proposed method to segment breast region, which is usually the first step of computer-aided system for breast mass detection in mammography. In experiment, we randomly select 100 images from the digital database for screening mammography (Heath et al., 2001). The ground truth for segmentation is labeled by radiologist in advance. We compare the proposed method and the other four methods mentioned above by the AOM (area overlap measure), which is defined as:

$$AOM = \frac{|A \cap T|}{|A \cup T|} \tag{13}$$

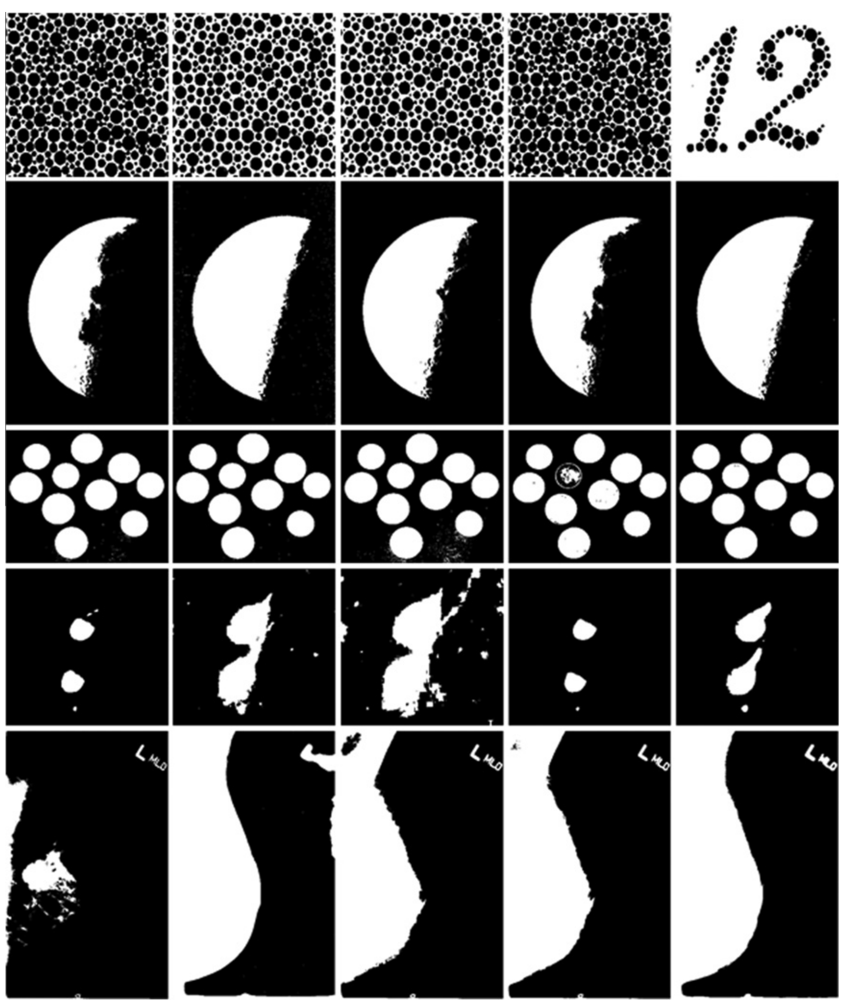

Fig. 4. Segmentation results. From left to right column: results of Kapur, Rosin, Medina-Carnicer, Otsu, and ours.

Table 2
The thresholds of the five segmentation methods.

Images	Kapur	Rosin	Medina-Carnicer	Otsu	Our method
Color vision image	124	82	85	125	47
Moon image	125	13	74	89	33
Coin image	78	79	76	126	88
Sperm image	85	28	22	103	42
Mammogram	185	32	112	99	53

Table 3
Descriptive statistics of area overlap measure for the five segmentation methods.

Method	Min.	1st Qu.	Med	3rd Qu.	Max	Mean	Std
Kapur	0.02	0.18	0.32	0.64	0.79	0.39	0.25
Rosin	0.44	0.80	0.89	0.93	0.96	0.84	0.12
Medina-Carnicer	0.18	0.65	0.71	0.75	0.85	0.69	0.09
Otsu	0.52	0.73	0.76	0.81	0.87	0.76	0.06
Our method	0.72	0.86	0.88	0.92	0.96	0.88	0.04

where A represents the automatically segmented region, T represents the ground truth, and $|.|$ represents the number of pixels in the set.

Table 3 shows the descriptive statistics of area overlap measure. The statistics demonstrate that our method is robust and can obtain good results for breast region segmentation in mammograms.

6. Conclusions

A property of Otsu method is discovered in this paper. It is that Otsu threshold is equal to the average of the mean levels of two classes partitioned by this threshold. We conclude from the property that Otsu threshold will bias in favor of the class with larger variance when the within-class variances of two classes are substantially different. Bases on this property, a novel range-constrained Otsu method is proposed for the case where the difference of the variances of object and background is significant. The experimental results on test images show that the effects of the proposed algorithm are satisfactory. The discovery in this paper can provide support for some improvements to Otsu method.

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